

Yash Patel

(289) 990 - 2434 | ytpatel@uwaterloo.ca | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Mechatronics Engineering

Sep. 2022 – May 2027

Relevant Courses: Intro to Thermodynamics & Heat Transfer, Computer Structures and RTOS, Data Structures and Algorithms, Design & Dynamics of Machines, Microprocessor Systems and Interfacing, Sensors and Instrumentation

EXPERIENCE

Autonomous Robotics SWE Intern

Sep. 2025 – Dec. 2025

Daylun Housing

Toronto, ON

- Designed and implemented a **multi-waypoint path-planning algorithm** for an AMR navigating sequenced assembly stations in a modular housing panel production line.
- Developed an **autonomous pick-and-place system** using **ROS2, MoveIt, and Gazebo**, writing custom joint-space controllers for stud placement onto panel frames, achieving **95%** placement accuracy in simulation.
- Engineered **closed-loop nailing, spray-foam dispensing, and steel-strapping subsystems**, integrating actuator control and sensor feedback to automate sequential panel fabrication steps.
- Simulated the **full AMR pipeline in NVIDIA IsaacSim** across **6** stations, implementing per-station **state machines** to validate end-to-end timing and correctness, reducing debugging cycles by **20%**.

Co-Founder & Erosion/Software Team Lead

Nov. 2023 – May 2025

WatDig Design Team

Austin, TX

- Developed a **React**-based GUI to subscribe to **ROS2** nodes and display real-time data from published sources, reducing monitoring latency by **30%** and enhancing situational awareness for autonomous vehicle operations.
- Implemented a **SLAM-based localization system** using **Python** and **Ultra WideBand (UWB)** technology to accurately track the robot's position relative to UWB anchor points, attaining a localization accuracy of **±10cm**.
- Designed the **motor, gearbox, and couplings** in **OnShape** to withstand the **2200 Nm** torque requirement.
- Utilized **rapid prototyping** to create enclosures and mounts for above and below ground PCB control stations.

CAD/CAM Programmer

Jan. 2025 – Apr. 2025

Sable Metal Manufacturing

Markham, ON

- Programmed sheet metal **bending sequences** using **INFORM II** for the **Motoman HP-50** robotic arm to automate production, resulting in a **15%** increase of efficiency.
- Engineered a novel **end-of-arm tooling system** incorporating a **pneumatic gripper** and suction cups to expand part-handling capabilities, supporting **3+** distinct part geometries and reducing changeover time by **25%**.
- Implemented a custom **mechanical limit switch** based on **parallel linkage mechanics** to enable seamless production across **5+** varying part geometries, reducing manual adjustments by **30%**.

Vehicle Platform Division - Interfacing/Mechanical Team Member

Jan. 2024 – Nov. 2024

WATonomous Design Team

Waterloo, ON

- Developed **ROS2** nodes to interface **Velodyne LiDAR** and **camera** sensor streams, publishing outputs to the autonomy stack across **5+** topics for downstream perception and planning modules.
- Designed and fabricated a **roof rack** in **SolidWorks** to structurally mount and expose **4 LiDAR sensors** and **3 cameras**, optimizing sensor placement for **360°** coverage.
- Engineered a compact **SolidWorks**-designed casing to house an **emergency stop** and rocker switch within a cup holder, ensuring accessible vehicle safety controls while maintaining a clean and space-efficient interior integration.

Mechanical/Manufacturing Engineering Intern

Sep. 2023 – Dec. 2023

Linamar Corporation

Guelph, ON

- Created engineering **drawings** and **models** of battery enclosures, subframes, sprockets, and wheel spindles using both **SolidWorks** and **AutoCAD**, contributing to structure and chassis development.
- Automated part-sorting workflows using **Excel VBA**, reducing manual processing time by **15%** and improving reliability for monthly production deadlines.

SKILLS

Programming Languages: C++, Python, RobotC, Java, Excel VBA, MATLAB, SQL, Ladder Logic, INFORM II

Tools, Frameworks & Libraries: Tensorflow, PLC, YOLOv8, JIRA, Git, Linux, ROS2, Verilog, Docker, STM32

CAD & Mechanical Concepts: SolidWorks, Inventor, AutoCAD, Onshape, SketchUp, 5S, GD&T, DFM, FEA